



Research Concept Note

Modules 1 & 2 Assignment • Research Masterclass for Clinicians

Instructions

Write a research concept note that applies what you have learned across Module 1 and Module 2. Your concept note must be based on data from the Demographic and Health Surveys (DHS) Program. Choose one country and survey year relevant to your research question.

Your concept note must be 3 pages long (Sections 1–6 below), plus a required Applied Data Interpretation Exercise annex. Every section listed below must appear in your submission as its own numbered heading.

Skills You Will Demonstrate

Module 1 • Weeks 1–6

- Framing a public health problem epidemiologically
- Selecting and justifying an appropriate study design
- Working with population-level survey data
- Calculating and interpreting measures of disease occurrence
- Calculating and interpreting measures of effect and impact

Module 2 • Weeks 7–12

- Classifying variables by data type and summarising them appropriately
- Applying probability and statistical distributions to study data
- Formulating and testing a statistical hypothesis
- Choosing and justifying a test for comparing means or proportions
- Calculating and interpreting a measure of association

Required Sections

1. Background & Problem Statement

Describe the health problem, its burden, and why it warrants investigation. Frame the issue epidemiologically, using Week 1–2 concepts.

2. Research Question & Objectives

State a clear, answerable research question and 2–3 specific objectives.

3. Study Design & Justification

Propose a study design (Week 6) and justify why it is appropriate for your question. Note its strengths and limitations relative to alternative designs.

4. Data Source & Key Variables

Name the DHS country survey and year you are using. List the key variables you will use and classify each by data type (Week 8).

Access DHS datasets: <https://dhsprogram.com/data/available-datasets.cfm> — select your country and survey, then register for access to the dataset.

5. Epidemiological Measures

State which measures of disease occurrence and/or effect (Weeks 3 & 5) you will calculate from your DHS dataset, and what each will mean in your study's context.

6. Statistical Analysis Plan

Describe your planned analysis: how you will summarise your variables (Week 8), what distributional assumptions apply (Week 9), the hypothesis you are testing (Week 10), the specific test you will use to compare groups (Week 11), and the measure of association you will report (Week 12). Justify each choice.

Annex. Applied Data Interpretation Exercise (Required, not counted in the 3-page limit)

Find one results table or figure relevant to your research question from either the DHS StatCompiler tool or a published DHS Final Report for your chosen country. Include it in your annex, then answer all four of the following:

- What statistic is being shown (e.g. a proportion, a mean, a rate), and what two groups are being compared?
- What hypothesis is this comparison testing?
- Based on the values shown (and confidence intervals or sample sizes, if given), does the difference between the groups appear meaningful? Explain your reasoning.
- Name one limitation of using this data source or this comparison.

Find a table or figure: <https://www.statcompiler.com/en/> (DHS StatCompiler, no download needed) or <https://dhsprogram.com/publications/publication-search.cfm> (DHS Final Reports).

Format & Submission

Length:	3 pages (Sections 1–6), excluding the annex and references
Format:	PDF, 12pt font, 1.5 line spacing, numbered sections matching this brief
Due Date:	Sunday 18 October 2026, 11:59pm
Submission:	Email your completed PDF to research@newlandsclinic.org.zw

Grading Rubric

Criterion	Section	Weight	Excellent	Satisfactory	Needs Improvement
Background & Problem Statement	Section 1	10%	Problem is clearly framed epidemiologically, with burden and relevance well established.	Problem is described but epidemiological framing or relevance is underdeveloped.	Problem statement is vague, unsupported, or missing epidemiological context.
Research Question & Objectives	Section 2	10%	Question is specific, answerable, and objectives align tightly with it.	Question and objectives are present but somewhat broad or loosely aligned.	Question is unclear, unanswerable, or objectives are missing/misaligned.
Study Design & Justification	Section 3	15%	Design is appropriate and well justified, with strengths/limitations meaningfully discussed.	Design is appropriate but justification is thin or limitations not addressed.	Design is inappropriate for the question or not justified.
Data Source & Key Variables	Section 4	15%	Correct DHS country survey and year are cited; variables are correctly and completely classified by type.	DHS dataset is identified but citation is incomplete; some variable classifications are inconsistent.	Data source is not a DHS dataset, is unclear, or variables are missing/misclassified.
Statistical Analysis Plan	Sections 5 & 6	20%	Tests and measures are correctly chosen and justified relative to data type, distribution, and question.	Analysis plan is broadly sound but one or more choices are unjustified or a poor fit.	Analysis plan is missing, incorrect, or mismatched to the data/question.
Applied Data Interpretation	Annex	20%	Table/figure is relevant and included; the statistic, comparison, and hypothesis	Table/figure is included; most elements are correctly identified, but the interpretation	Table/figure is missing or unrelated, or the statistic,

Criterion	Section	Weight	Excellent	Satisfactory	Needs Improvement
			are correctly identified; the interpretation of meaningfulness and the limitation are accurate and well explained.	or limitation is underdeveloped.	hypothesis, and interpretation are incorrect or absent.
Clarity, Structure & Scientific Writing	Overall	10%	Well organised, concise, and follows scientific writing conventions throughout.	Generally clear but with some structural or stylistic inconsistencies.	Difficult to follow; poorly structured or does not follow scientific conventions.
Total		100%			

If you need a refresher on any topic referenced above, revisit the relevant week on the Course Material page before you start writing.